

Why and What for (Four): The Basis for Writing a Good Introduction

by Emmanuel P. Papadakis,*
Research Coordinator, *Materials Evaluation*

INTRODUCTION

The systematic writing of the introduction of a paper is important because the introduction outlines several important aspects of the paper and because the writing process focuses the thoughts of the author upon the logical organization of his presentation. Most scientists and engineers have received little formal training in writing and write introductions by a "seat-of-the-pants" method that involves mostly intuitive or experiential feelings as to the proper contents and sequence, causing some items to be left out or put out of order. While there is no hard-and-fast rule for writing good introductions, the author has discovered a sequence of five topics that will be helpful to most authors of technical papers. The author has attempted to describe in a systematic fashion the requirements for a good introduction, to list them in a memorable way, and to illustrate their use. The present short note describes the five-point approach and gives some examples of its applications.

BACKGROUND

"Well begun is half done," reads the old English proverb. This adage holds true in the writing of technical papers. Once the scientist or engineer has completed his technical work and sits down to face a pad of paper or a word processor, his work begins in earnest. He has plenty to write about. He has an urge to tell the whole world. Yet how to begin? He ponders, sips coffee, squirms in his chair, smokes nervously, and ponders more. Technology is his forte, not liter-

ature. It was rough in college; it is even worse now.

Writing a good introduction is the first step, the "well begun" part of the proverb. Once this is accomplished, the paper does *seem* half done. There may be reams of writing under other sections, such as theory, experiment, results, and discussion, but the introduction is the initial obstacle to overcome if the rest of the paper is to fall into place.

The introduction ought to inform the audience about the author's effort. Certainly, the work was worth doing, wasn't it? Why should the author want to tell me (the audience) about it? Why should I want to read about it? Even if he and I have a personal interest in his subject, why is it worth printing in the archival literature? What is the reader to look for as he reads onward (or decides not to)? The introduction begins to answer these questions, serves to orient the reader, and attempts to convince him to read further. It is the salesman's "foot in the door" translated into literary terms. When written effectively, it also organizes the thoughts of the author so he can write more effectively and complete his task more quickly.

THEORY

To be effective, the introduction should answer the questions "Why and What For (Four)?" Expanded, these questions are:

- Why is the topic of interest?
- What (1) is the background on the previous solutions, if any?
- What (2) is the background on potential solutions?
- What (3) was attempted in the present effort (research project)?
- What (4) will be presented in this paper?

This mnemonic format really works. The author formalized it by analyzing

the introduction to a manuscript written by another ASNT member. That introduction was so logical and complete that it seemed worth summarizing for wider use. The resulting summary was the mnemonic, "Why and What For (Four)," cited above. After twenty years of writing introductions that were incomplete, out of order, verbose, and probably less effectual than they should have been, the author has adopted the above format and finds it very useful.

EXAMPLES

The following are very short examples of introductions written by using the "Why and What For (Four)" method for hypothetical technical papers. Any correspondence between these and the output of real people is purely coincidental. However, by using exaggerated examples, the principles are illustrated.

Ultrasonic Transducers with No Beam-Spreading, by P. Z. T. Backing

Why is the topic of interest? Ultrasonic transducers with no beam-spreading and constant amplitude and phase profiles within the beam would be useful in studies of flaw detection and materials properties. Quantitative non-destructive evaluation could be done expeditiously because the wave impinging upon a flaw would be exactly known and very simple in form. Ultrasonic attenuation and velocity measurements could be made in all sizes of specimens without recourse to beam-spreading (diffraction) corrections and without interference from sidewall boundaries.

What is the background on the previous solutions, if any? Up until the present time, available ultrasonic transducers have produced beam spreading of both the central lobe and the geometrical side lobes of energy. The

*Manufacturing Processes Laboratory, Ford Motor Company, 24500 Glendale Ave., Detroit, MI 48239

most advanced laboratory techniques to date have produced a Gaussian central lobe that spreads without side lobes. None of the present transducers can produce constant amplitude and phase fronts within the beam.

What is the background on potential solutions? With the advent of microelectronics, the possibility has arisen to build a phased array transducer on a piezoelectric macrochip using a three-dimensional, multi-layer printed circuit for a backing. A small handheld transducer with a 1000×1000 array of radiators and 10 circuit elements per array element should be capable of producing the desired phase and amplitude effects.

What was attempted in the present effort (research project)? In the present study, the theory was worked out for the proposed transducer and also for a 10×10 array pilot model. This model was built and tested experimentally. The pilot model was evaluated by several electronic and optical methods. Three critical experiments were then performed on quantitative flaw evaluation, ultrasonic attenuation, and ultrasonic velocity, respectively.

What will be presented in this paper? In this paper, the theory will be presented for both arrays. Construction details of the pilot model will be given. The most relevant results of its evaluation will be shown, and projections will be made concerning the expected improved performance of the 1000×1000 array. The results of the three experiments will also be enumerated, and comparisons will be made with results found using presently available transducers. Some fundamental limitations will be pointed out.

A New Method for Transporting Potable Water, by Aquus Transductus IV

Why is the topic of interest? Po-

table water is in short supply in several cities while it is abundant in the adjacent provinces, particularly in the mountains not far distant. It would be advantageous to transport the available water from the locus of supply to the locus of need without the continual burden on human and animal transport systems presently employed.

What is the background on the previous solutions, if any? Previous facilities such as wells and cisterns are proving inadequate for the urban population. Ox carts with large barrels are an interim measure, but they are of limited utility because they overburden the roads and cannot be relied upon in seasons when the oxen are needed for farming. Terra cotta pipe, used first in Crete by the Minoans, is usable only for short distances. It cannot withstand the hydrostatic pressure in a valley when the line of flow is down one hill and up another. Previous proposals for elevated aqueducts have been rejected as infeasible because of the short span bridged by a single lintel stone. See, for instance, the width of the gateway of the Temple of Agamemnon and the number of columns needed in the Parthenon.

What is the background on potential solutions? The proposed solution is an improved elevated aqueduct based on a new invention—the "arch." The "arch" is a structure of fitted stones describing a circle in the vertical plane. It can span a gap at least one order of magnitude longer than is possible with the largest practical lintel stone. Masonry structures, which can support great loads, can be built on top of the arch. The load-bearing capability of an arch is derived from the exact angles of the faces of the stones within the circle. The resolved forces from the load above and including the arch are thus transmitted perpendicularly to the faces of the stones, resulting in a stable structure because

there are no shearing forces on the joints and essentially no flexural forces on any stone, the nemesis of a lintel. The arches can be built atop columns, so the gentle slope of an aqueduct can be maintained above variable terrain with only $1/X$ as many columns as needed previously. This should result in a cost saving, making the new design feasible.

What was attempted in the present effort (research project)? In the present effort, an experimental aqueduct, XVIII stadia long, was built from the south branch of the North River to the eastern cistern of Westium. The cistern was enlarged, and control gates were constructed. The facilities were manned by the MCMXVI Engineers. This having been done, the aqueduct was operated for XIV months.

What will be presented in this paper? This paper will give the theory of the "arch," the method of calculating the angles of the faces of the stones including the "keystone," and the details of the masonry construction. Also, flow data for the rainy season and the dry season will be presented, demonstrating successful operation.

CONCLUSION

The "Why and What For (Four)" method of writing introductions to technical papers can be applied to most subjects. The reader will note that the first paragraph of this article contains five sentences that apply the five points of the method to the present subject—writing introductions. The two tongue-in-cheek introductions shown are more extensive examples. If authors choose to use the "Why and What For (Four)" method, they and their readers will reap the benefits of clarity, conciseness, simplicity, and completeness.